AS1109 - Late Mid-Sem Examination

Complete Student Solutions

Question 1

Q1(a)

Suppose $f(x,y) \to L$ as $(x,y) \to (0,0)$ along every straight line through the origin. Does it follow that

$$\lim_{(x,y)\to(0,0)} f(x,y) = L?$$

Justify.

Solution

No. Checking all straight-line paths is not sufficient for a two-variable limit. A non-linear path may give a different value.
Consider

$$f(x,y) = \begin{cases} \dfrac{x^2 y}{x^4+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Take any non-vertical straight line through the origin:

$$y = mx.$$

Then

$$f(x,mx) = \frac{x^2(mx)}{x^4+m^2x^2} = \frac{mx}{x^2+m^2}.$$

If $m \neq 0$, then as $x \to 0$,

$$\frac{mx}{x^2+m^2} \to 0.$$

If $m = 0$, then $y = 0$ and

$$f(x,0) = 0.$$

On the vertical line $x = 0$,

$$f(0,y) = 0.$$

Thus the limit along *every straight line* through the origin is 0.
Now approach along the parabola

$$y = x^2.$$

Then

$$f(x,x^2) = \frac{x^2(x^2)}{x^4+x^4} = \frac{1}{2} \qquad (x \neq 0).$$

Therefore the value along the parabola tends to $1/2$, not 0.
Hence

equal limits along all straight lines do not guarantee the full two-variable limit.

Key idea

To prove that a multivariable limit *does not exist*, it is enough to find two paths giving different limiting values. To prove that a limit *does exist*, checking many paths is not enough; one needs an argument that works for all approaches.

Q1(b)

If f_x and f_y exist at a point, is $f(x,y)$ necessarily continuous there? Give justification.

Solution

No. The existence of the two first-order partial derivatives at a point does not by itself guarantee continuity.

Consider

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

First compute the partial derivative with respect to x at the origin:

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h}.$$

Since

$$f(h,0) = 0,$$

we obtain

$$f_x(0,0) = 0.$$

Similarly,

$$f_y(0,0) = \lim_{k\to 0} \frac{f(0,k) - f(0,0)}{k} = 0.$$

So both partial derivatives exist.
However, along the line $y = x$,

$$f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2} \qquad (x \neq 0).$$

Hence

$$\lim_{x\to 0} f(x,x) = \frac{1}{2},$$

whereas

$$f(0,0) = 0.$$

Thus f is not continuous at the origin.
Therefore

$$\boxed{f_x \text{ and } f_y \text{ may both exist at a point while } f \text{ is not continuous there.}}$$

Q1(c)

Find all local maxima, local minima, and saddle points, if any, of

$$f(x,y) = e^{2x} + y^2.$$

Solution

The function is differentiable everywhere. Therefore any interior local maximum, local minimum, or saddle point must occur at a critical point.
Compute the first partial derivatives:

$$f_x(x,y) = 2e^{2x}, \qquad f_y(x,y) = 2y.$$

A critical point must satisfy

$$f_x = 0 \qquad \text{and} \qquad f_y = 0.$$

But

$$e^{2x} > 0$$

for every real x. Hence

$$2e^{2x} > 0$$

for every real x, so

$$f_x(x,y) \neq 0$$

everywhere.
Therefore the function has no critical points.
Hence

$$\boxed{\text{there are no local maxima, no local minima, and no saddle points.}}$$

Question 2

Q2

For

$$f(x, y) = xy,$$

find the range of f and sketch the level curves

$$f(x, y) = c$$

for $c = 0, 1$.

Solution

We first find the range.
Let $r \in \mathbb{R}$ be arbitrary. Choose

$$x = r, \qquad y = 1.$$

Then

$$f(r, 1) = r.$$

Thus every real number occurs as a value of f. Hence

$$\boxed{\text{Range}(f) = \mathbb{R}.}$$

Now find the level curves.
For $c = 0$,

$$xy = 0.$$

A product is zero when at least one factor is zero, so

$$x = 0 \qquad \text{or} \qquad y = 0.$$

Thus the level curve is

$$\boxed{\text{the union of the } x\text{-axis and } y\text{-axis.}}$$

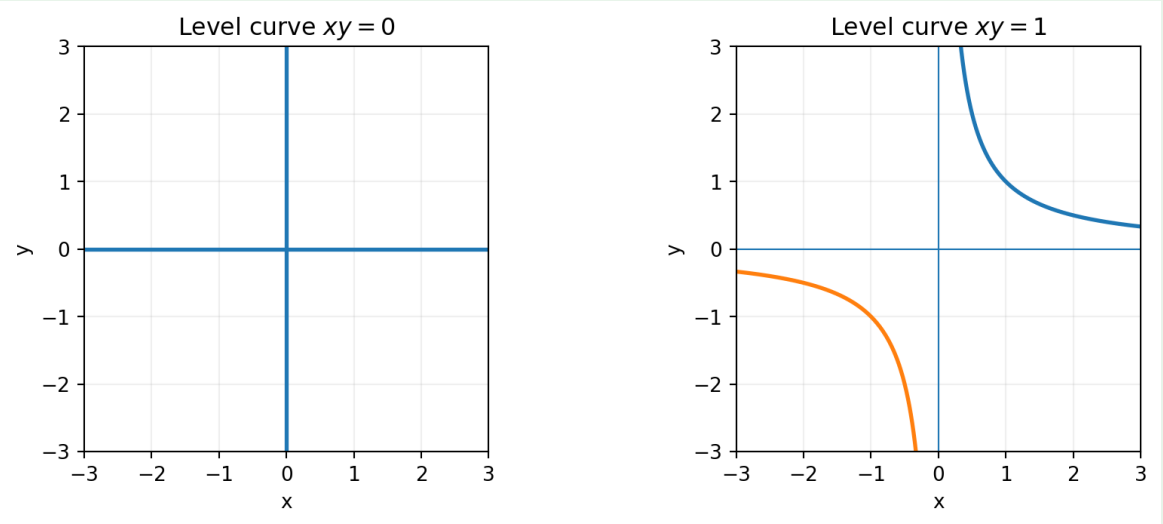
For $c = 1$,

$$xy = 1.$$

Therefore

$$y = \frac{1}{x}, \qquad x \neq 0.$$

This is a rectangular hyperbola with one branch in Quadrant I and the other in Quadrant III.



Question 3

Q3(a)

Evaluate, if it exists,

$$\lim_{(x,y)\to(0,0)} \frac{x(x^2-y^2)}{x^2+y^2}.$$

Solution using polar coordinates

Use

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad x^2+y^2 = r^2.$$

Also,

$$x^2 - y^2 = r^2(\cos^2\theta - \sin^2\theta) = r^2\cos 2\theta.$$

Therefore

$$\frac{x(x^2-y^2)}{x^2+y^2} = \frac{(r\cos\theta)(r^2\cos 2\theta)}{r^2} = r\cos\theta\cos 2\theta.$$

Since

$$|\cos\theta| \le 1, \qquad |\cos 2\theta| \le 1,$$

we have

$$|r\cos\theta\cos 2\theta| \le r.$$

As $(x,y)\to(0,0)$, we have $r\to 0$. Hence, by the squeeze theorem,

$$\boxed{\lim_{(x,y)\to(0,0)} \frac{x(x^2-y^2)}{x^2+y^2} = 0.}$$

A shorter inequality check

Since

$$|x^2 - y^2| \le x^2 + y^2,$$

we also have

$$\left|\frac{x(x^2-y^2)}{x^2+y^2}\right| \le |x| \to 0.$$

So the same result follows directly.

Q3(b)

Evaluate

$$\lim_{(x,y)\to(2,2)} \frac{x+y-4}{\sqrt{x+y}-2}, \qquad x+y \neq 4.$$

Solution

As $(x,y)\to(2,2)$,

$$x+y \to 4.$$

Direct substitution gives the indeterminate form

$$\frac{0}{0}.$$

Rationalise the denominator:

$$\frac{x+y-4}{\sqrt{x+y}-2}\cdot\frac{\sqrt{x+y}+2}{\sqrt{x+y}+2}.$$

Since

$$x+y-4 = (\sqrt{x+y}-2)(\sqrt{x+y}+2),$$

and $x+y\neq 4$, we may cancel:

$$\frac{x+y-4}{\sqrt{x+y}-2} = \sqrt{x+y}+2.$$

Therefore

$$\lim_{(x,y)\to(2,2)} \frac{x+y-4}{\sqrt{x+y}-2} = \sqrt{4}+2 = \boxed{4}.$$

Question 4

Q4

For

$$h(x,y) = \frac{\ln(1-x^2-y^2)}{x-y},$$

(a) find and sketch the domain;

(b) find the points in $\mathbb{R}^2$ where the function is continuous.

Solution

There are two restrictions.
Restriction 1: the logarithm.
For $\ln(1-x^2-y^2)$ to be defined,

$$1-x^2-y^2 > 0.$$

Therefore

$$x^2+y^2 < 1.$$

This is the *open* unit disk.
Restriction 2: the denominator.
We require

$$x-y \neq 0,$$

so

$$x \neq y.$$

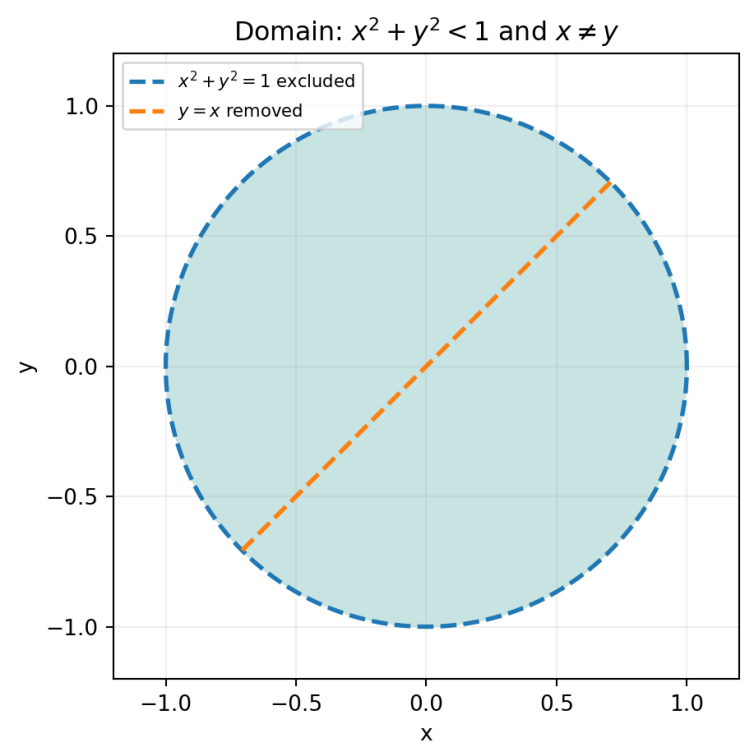
Hence the domain is

$$\boxed{D = \{(x,y) \in \mathbb{R}^2 : x^2+y^2 < 1,\ x \neq y\}.}$$

Geometrically, start with the open unit disk and remove the part of the line

$$y = x$$

lying inside the disk.



Now consider continuity.
The function

$$1-x^2-y^2$$

is continuous everywhere. The logarithm is continuous where its argument is positive, so

$$\ln(1 - x^2 - y^2)$$

is continuous whenever

$$x^2 + y^2 < 1.$$

The denominator

$$x - y$$

is continuous everywhere and is nonzero precisely when

$$x \neq y.$$

A quotient of continuous functions is continuous wherever its denominator is nonzero. Therefore h is continuous at every point of its domain:

$$\boxed{h \text{ is continuous exactly at } \{(x, y) : x^2 + y^2 < 1,\ x \neq y\}.}$$

Question 5

Q5

Let

$$z = x^2 + y^2, \qquad x = s + t, \qquad y = s - t.$$

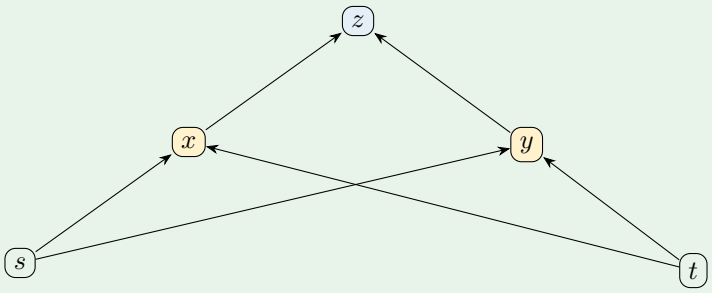
Using the chain rule, find

$$\frac{\partial z}{\partial s} \quad \text{and} \quad \frac{\partial z}{\partial t}$$

in terms of the independent variables s, t.

Solution

The dependence is



First,

$$z_x = 2x, \qquad z_y = 2y.$$

Also,

$$x_s = 1, \qquad y_s = 1,$$

and

$$x_t = 1, \qquad y_t = -1.$$

By the multivariable chain rule,

$$\frac{\partial z}{\partial s} = z_x x_s + z_y y_s.$$

Thus

$$\frac{\partial z}{\partial s} = 2x + 2y.$$

Substitute

$$x = s + t, \qquad y = s - t:$$

$$\frac{\partial z}{\partial s} = 2(s + t) + 2(s - t) = \boxed{4s}.$$

Similarly,

$$\frac{\partial z}{\partial t} = z_x x_t + z_y y_t.$$

Therefore

$$\frac{\partial z}{\partial t} = 2x - 2y.$$

Substitute $x = s + t,\ y = s - t$:

$$\frac{\partial z}{\partial t} = 2(s + t) - 2(s - t) = \boxed{4t}.$$

Direct-substitution check

$$z = (s + t)^2 + (s - t)^2 = 2s^2 + 2t^2.$$

Therefore

$$z_s = 4s, \qquad z_t = 4t,$$

confirming the chain-rule result.

Question 6

Q6

Let

$$w = f(u) + g(v), \qquad u = x + 2y, \qquad v = x - 2y.$$

Show that

$$w_{xx} - \frac{1}{4}w_{yy} = 0.$$

Solution

We differentiate using the chain rule.
Because

$$u_x = 1, \qquad v_x = 1,$$

we have

$$w_x = f'(u)u_x + g'(v)v_x = f'(u) + g'(v).$$

Differentiate once more with respect to x:

$$w_{xx} = f''(u)u_x + g''(v)v_x.$$

Therefore

$$\boxed{w_{xx} = f''(u) + g''(v)}.$$

Now differentiate with respect to y.
Since

$$u_y = 2, \qquad v_y = -2,$$

we get

$$w_y = 2f'(u) - 2g'(v).$$

Differentiate again with respect to y:

$$w_{yy} = 2f''(u)u_y - 2g''(v)v_y.$$

Hence

$$w_{yy} = 2f''(u)(2) - 2g''(v)(-2) = 4f''(u) + 4g''(v).$$

Thus

$$\frac{1}{4}w_{yy} = f''(u) + g''(v).$$

Comparing this with w_{xx},

$$w_{xx} = \frac{1}{4}w_{yy}.$$

Therefore

$$\boxed{w_{xx} - \frac{1}{4}w_{yy} = 0.}$$

Question 7

Q7

Find the linear approximation $L(x, y, z)$ to

$$f(x, y, z) = e^x + \sin(y + z)$$

at

$$P = \left(0, \frac{\pi}{2}, 0\right).$$

Solution

For a differentiable function of three variables, the linear approximation at

$$(a, b, c)$$

is

$$L(x,y,z) = f(a,b,c) + f_x(a,b,c)(x-a) + f_y(a,b,c)(y-b) + f_z(a,b,c)(z-c).$$

Here,

$$(a,b,c) = \left(0, \frac{\pi}{2}, 0\right).$$

First evaluate f:

$$f\left(0, \frac{\pi}{2}, 0\right) = e^0 + \sin\frac{\pi}{2} = 1 + 1 = 2.$$

Now compute the partial derivatives:

$$f_x = e^x,$$
$$f_y = \cos(y+z),$$
$$f_z = \cos(y+z).$$

At the given point,

$$f_x(P) = 1, \qquad f_y(P) = 0, \qquad f_z(P) = 0.$$

Therefore

$$\begin{aligned} L(x,y,z) &= 2 + 1(x-0) + 0\left(y - \frac{\pi}{2}\right) + 0(z-0) \\ &= \boxed{x+2}. \end{aligned}$$

Question 8

Q8

In a thin-lens model,

$$F(u,v) = \frac{uv}{u+v},$$

where u is the object distance and v is the image distance.
At a particular setting,

$$u = 20 \text{ cm}, \qquad v = 30 \text{ cm}.$$

Each distance may have a measurement error of at most 0.1 cm.

(a) Find F_u and F_v, and write the total differential dF.

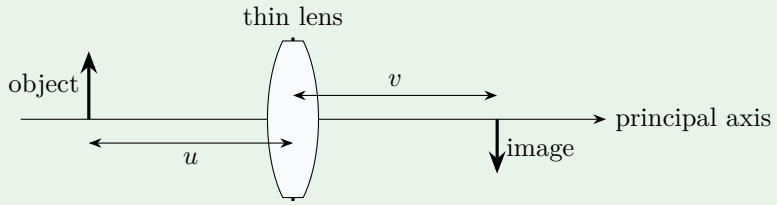
(b) Use differentials to estimate the maximum absolute error in the computed focal length.

(c) Which measurement contributes more to the first-order uncertainty in F? Justify.

Solution

The model is

$$F(u,v) = \frac{uv}{u+v}.$$



(a) Partial derivatives and total differential
Using the quotient rule,

$$F_u = \frac{v(u+v) - uv}{(u+v)^2} = \frac{v^2}{(u+v)^2}.$$

Similarly,

$$F_v = \frac{u(u+v) - uv}{(u+v)^2} = \frac{u^2}{(u+v)^2}.$$

Hence

$$\boxed{dF = \frac{v^2}{(u+v)^2}\,du + \frac{u^2}{(u+v)^2}\,dv.}$$

At

$$u = 20, \qquad v = 30,$$

we have

$$u + v = 50.$$

Therefore

$$F_u = \frac{30^2}{50^2} = \frac{900}{2500} = \frac{9}{25} = 0.36,$$

and

$$F_v = \frac{20^2}{50^2} = \frac{400}{2500} = \frac{4}{25} = 0.16.$$

Thus, near the given setting,

$$\boxed{dF \approx 0.36\,du + 0.16\,dv.}$$

(b) Maximum absolute error
The measurement errors satisfy

$$|du| \le 0.1, \qquad |dv| \le 0.1.$$

Using the differential estimate,

$$|dF| \le |F_u||du| + |F_v||dv|.$$

Therefore

$$|dF| \le 0.36(0.1) + 0.16(0.1).$$

So

$$|dF| \leq 0.036 + 0.016 = 0.052.$$

Hence the estimated maximum absolute error is

$$\boxed{0.052 \text{ cm.}}$$

(c) Which measurement contributes more?
The first-order contribution from the u-measurement is

$$|F_u||du| \leq 0.36(0.1) = 0.036 \text{ cm.}$$

The contribution from the v-measurement is

$$|F_v||dv| \leq 0.16(0.1) = 0.016 \text{ cm.}$$

Since

$$0.036 > 0.016,$$

the object-distance measurement contributes more.
Therefore

$$\boxed{u \text{ contributes more to the first-order uncertainty in } F.}$$

Question 9

Q9

Consider

$$f(x,y) = x^4 + y^4 - 4xy.$$

(a) Find all the critical points.

(b) Compute

$$D = f_{xx}f_{yy} - (f_{xy})^2$$

at every critical point.

(c) Classify each critical point as a local maximum, local minimum, or saddle point.

Solution

(a) Critical points

Compute the first partial derivatives:

$$f_x = 4x^3 - 4y,$$
$$f_y = 4y^3 - 4x.$$

At a critical point,

$$f_x = 0, \qquad f_y = 0.$$

Thus

$$4x^3 - 4y = 0 \implies y = x^3,$$

and

$$4y^3 - 4x = 0 \implies x = y^3.$$

Substitute $y = x^3$ into $x = y^3$:

$$x = (x^3)^3 = x^9.$$

Hence

$$x^9 - x = 0,$$

so

$$x(x^8 - 1) = 0.$$

For real x,

$$x = 0, \quad x = 1, \quad x = -1.$$

Since $y = x^3$, the corresponding y-values are

$$0, \quad 1, \quad -1.$$

Therefore the critical points are

$$\boxed{(0,0),\ (1,1),\ (-1,-1).}$$

(b) Compute D

The second derivatives are

$$f_{xx} = 12x^2, \qquad f_{yy} = 12y^2, \qquad f_{xy} = -4.$$

Hence

$$D = (12x^2)(12y^2) - (-4)^2 = 144x^2y^2 - 16.$$

At $(0,0)$,

$$D = -16 < 0.$$

At $(1,1)$,

$$D = 144 - 16 = 128 > 0.$$

At $(-1,-1)$,

$$D = 144 - 16 = 128 > 0.$$

(c) Classification

The second derivative test says:

$D < 0$	saddle point
$D > 0,\ f_{xx} > 0$	local minimum
$D > 0,\ f_{xx} < 0$	local maximum

At $(0,0)$,

$$D < 0,$$

so

$$\boxed{(0,0) \text{ is a saddle point.}}$$

At $(1,1)$,

$$D > 0 \qquad \text{and} \qquad f_{xx} = 12 > 0,$$

so

$$\boxed{(1,1) \text{ is a local minimum.}}$$

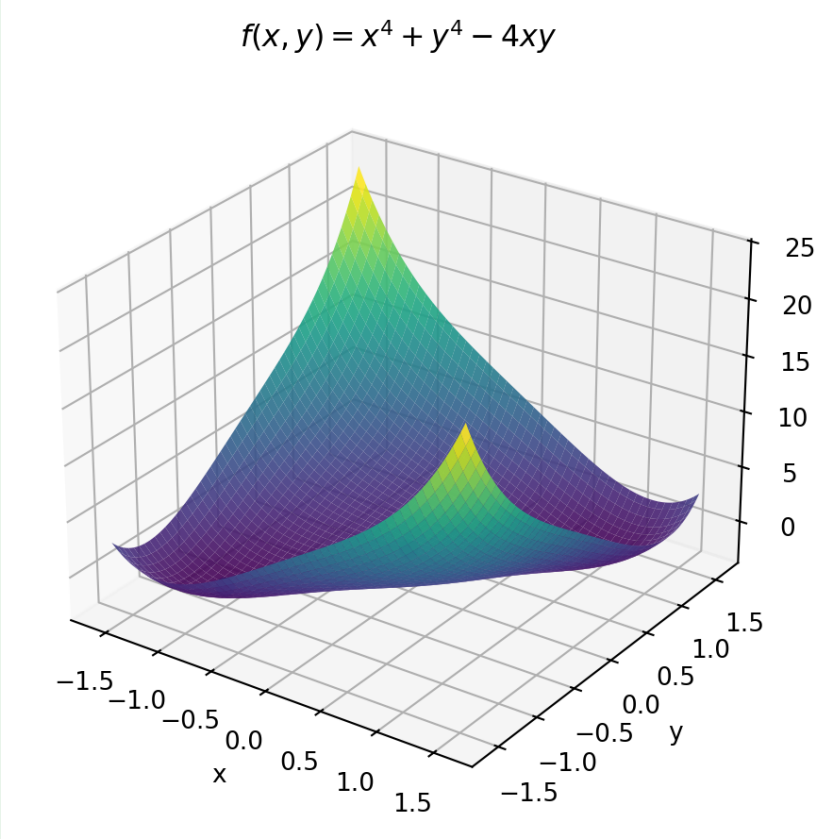
At $(-1,-1)$,

$$D > 0 \qquad \text{and} \qquad f_{xx} = 12 > 0,$$

so

$$\boxed{(-1,-1) \text{ is a local minimum.}}$$

There are no local maxima.



Thus

$(0,0)$	saddle point
$(1,1)$	local minimum
$(-1,-1)$	local minimum

Question 10

Q10(a)

Find the length of the curve

$$y = \int_{-2}^{x} \sqrt{3t^4 - 1}\, dt$$

from $x = -2$ to $x = -1$.

Solution

For a curve $y = y(x)$, the arc length from $x = a$ to $x = b$ is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

Here

$$y(x) = \int_{-2}^{x} \sqrt{3t^4 - 1}\, dt.$$

By the Fundamental Theorem of Calculus,

$$\frac{dy}{dx} = \sqrt{3x^4 - 1}.$$

Therefore

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + (3x^4 - 1) = 3x^4.$$

Hence

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{3x^4} = \sqrt{3}\, x^2,$$

because $x^2 \geq 0$.
Thus

$$L = \sqrt{3} \int_{-2}^{-1} x^2\, dx.$$

Therefore

$$\begin{aligned} L &= \sqrt{3} \left[\frac{x^3}{3}\right]_{-2}^{-1} \\ &= \frac{\sqrt{3}}{3} \left((-1)^3 - (-2)^3\right) \\ &= \frac{\sqrt{3}}{3}(-1 + 8) \\ &= \boxed{\frac{7\sqrt{3}}{3}}. \end{aligned}$$

Q10(b)

Find a curve through the point $(0, 1)$ whose length integral is

$$L = \int_{-\pi/2}^{\pi/2} \sqrt{1 + \sin^2 x}\, dx.$$

How many such curves are there? Give reason.

Solution

For a graph $y = y(x)$, the arc-length integrand is

$$\sqrt{1 + (y')^2}.$$

We are given

$$\sqrt{1 + (y')^2} = \sqrt{1 + \sin^2 x}.$$

Squaring both sides gives

$$(y')^2 = \sin^2 x.$$

Hence

$$y' = \pm \sin x.$$

First choose

$$y' = -\sin x.$$

Integrating,

$$y = \cos x + C.$$

The curve passes through $(0, 1)$, so

$$1 = \cos 0 + C = 1 + C.$$

Thus

$$C = 0,$$

and one curve is

$$\boxed{y = \cos x}.$$

Alternatively, choose

$$y' = \sin x.$$

Then

$$y = -\cos x + C.$$

Using $y(0) = 1$,

$$1 = -1 + C,$$

so

$$C = 2.$$

Thus another curve is

$$\boxed{y = 2 - \cos x}.$$

There is one further subtlety. On the interval

$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

$\sin x = 0$ only at $x = 0$. In the usual C^1 arc-length setting, the sign in

$$y' = \pm \sin x$$

may be chosen independently on the left and right of 0, because both choices give

$$y'(0) = 0.$$

Therefore the four C^1 curves through $(0, 1)$ are

$$\boxed{y = \cos x}$$

$$\boxed{y = 2 - \cos x}$$

$$\boxed{y = \begin{cases} \cos x, & x \leq 0, \\ 2 - \cos x, & x \geq 0, \end{cases}}$$

and

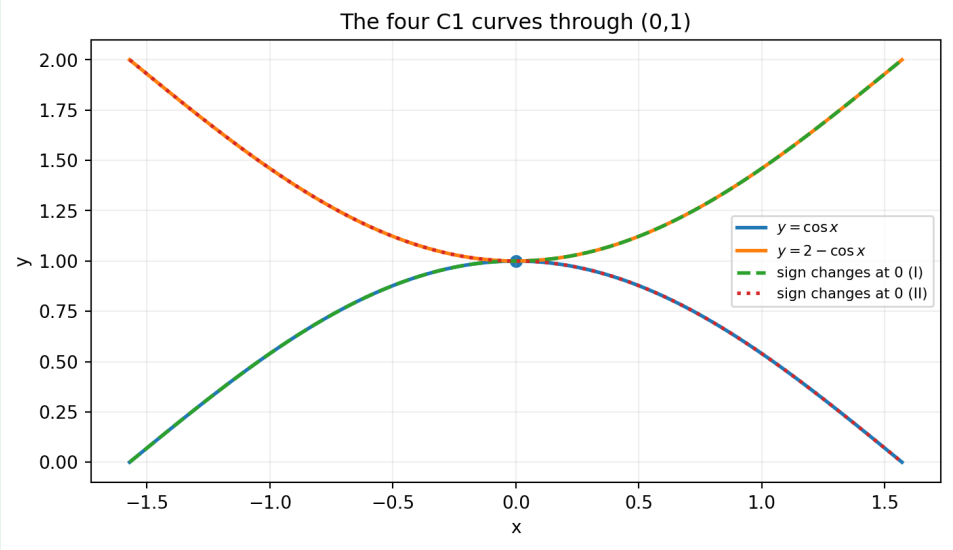
$$\boxed{y = \begin{cases} 2 - \cos x, & x \leq 0, \\ \cos x, & x \geq 0. \end{cases}}$$

All four satisfy

$$(y')^2 = \sin^2 x$$

and pass through $(0, 1)$. Therefore each has the required length integral.
Hence

$$\boxed{\text{there are four such } C^1 \text{ curves on the stated interval.}}$$



Why there are no additional C^1 choices

Away from $x = 0$, $\sin x \neq 0$. A continuous derivative cannot switch between $+\sin x$ and $-\sin x$ without passing through 0. Thus the sign is fixed on each side of 0, giving exactly

$$2 \times 2 = 4$$

choices.